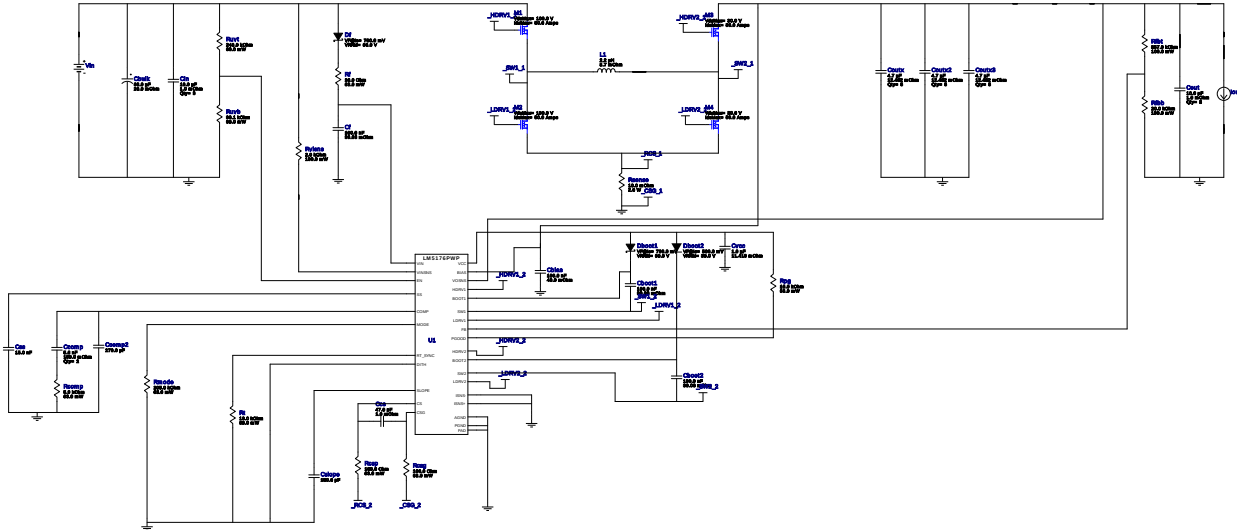
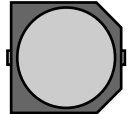


WEBENCH® Design Report

Design : 93 LM5176PWPR
LM5176PWPR 5V-35V to 15.00V @ 3A

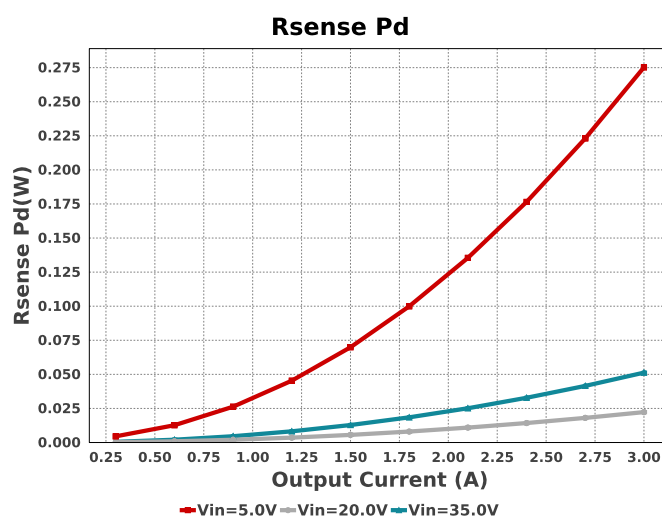
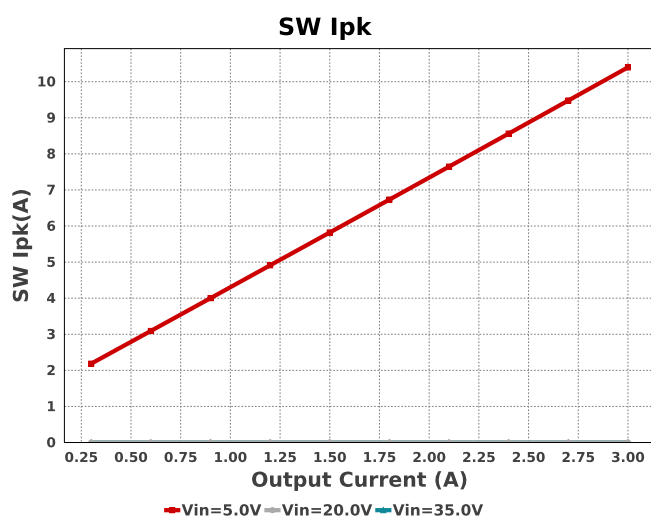


Electrical BOM

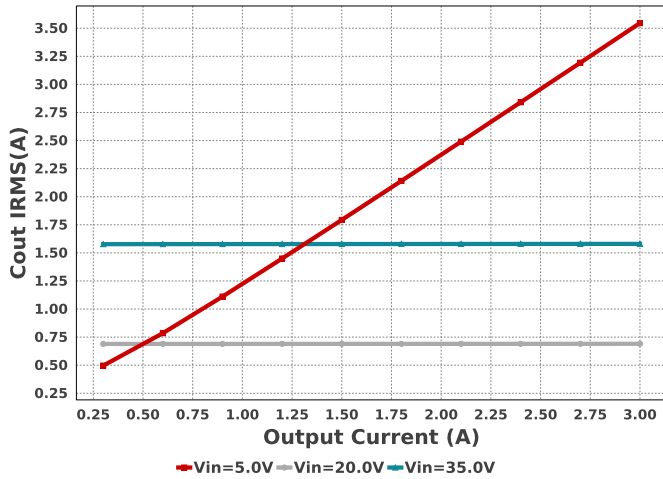
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cbias	TDK	C1005X5R1E104K050BC Series= X5R	Cap= 100.0 nF ESR= 40.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	\$0.02	0402 3 mm ²
Cboot1	TDK	C1005X7R1H104K050BB Series= X7R	Cap= 100.0 nF ESR= 39.03 mOhm VDC= 50.0 V IRMS= 0.0 A	1	\$0.02	0402 3 mm ²
Cboot2	TDK	C1005X7R1H104K050BB Series= X7R	Cap= 100.0 nF ESR= 39.03 mOhm VDC= 50.0 V IRMS= 0.0 A	1	\$0.02	0402 3 mm ²
Cbulk	Panasonic	50SVPF68M Series= SVPF	Cap= 68.0 µF ESR= 20.0 mOhm VDC= 50.0 V IRMS= 4.3 A	1	\$1.57	 CAPSMT_62_F12 151 mm ²
Ccomp	Kemet	C0805C562K5RACTU Series= X7R	Cap= 5.6 nF ESR= 199.0 mOhm VDC= 50.0 V IRMS= 297.0 mA	2	\$0.03	0805 7 mm ²
Ccomp2	Taiyo Yuden	UMK105CG271JV-F Series= C0G/NP0	Cap= 270.0 pF VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	0402 3 mm ²
Ccs	MuRata	GRM1555C1H470JA01D Series= C0G/NP0	Cap= 47.0 pF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	0402 3 mm ²
Cf	TDK	C1005X7R1H104K050BB Series= X7R	Cap= 100.0 nF ESR= 39.03 mOhm VDC= 50.0 V IRMS= 0.0 A	1	\$0.02	0402 3 mm ²

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cin	TDK	C3225X7R1H106M250AC Series= X7R	Cap= 10.0 uF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 5.0 A	3	\$0.27	 1210 15 mm ²
Cout	TDK	C3225X7R1H106M250AC Series= X7R	Cap= 10.0 uF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 5.0 A	3	\$0.27	 1210 15 mm ²
Coutx	Taiyo Yuden	MSASG219LB5475MTNA01 Series= X5R	Cap= 4.7 uF ESR= 12.482 mOhm VDC= 35.0 V IRMS= 1.66144 A	5	\$0.06	 0805 7 mm ²
Coutx2	Taiyo Yuden	MSASG219LB5475MTNA01 Series= X5R	Cap= 4.7 uF ESR= 12.482 mOhm VDC= 35.0 V IRMS= 1.66144 A	5	\$0.06	 0805 7 mm ²
Coutx3	Taiyo Yuden	MSASG219LB5475MTNA01 Series= X5R	Cap= 4.7 uF ESR= 12.482 mOhm VDC= 35.0 V IRMS= 1.66144 A	5	\$0.06	 0805 7 mm ²
Cslope	Taiyo Yuden	UMK105CG151JV-F Series= C0G/NP0	Cap= 150.0 pF VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	 0402 3 mm ²
Css	Kemet	C0603C153J3GACTU Series= C0G/NP0	Cap= 15.0 nF VDC= 25.0 V IRMS= 0.0 A	1	\$0.09	 0603 5 mm ²
Cvcc	TDK	C1005X6S1C105K050BC Series= X6S	Cap= 1.0 uF ESR= 11.416 mOhm VDC= 16.0 V IRMS= 1.483 A	1	\$0.02	 0402 3 mm ²
Dboot1	Diodes Inc.	B260A-13-F	VF@Io= 700.0 mV VRRM= 60.0 V	1	\$0.10	 SMA 37 mm ²
Dboot2	Diodes Inc.	B230A-13-F	VF@Io= 500.0 mV VRRM= 30.0 V	1	\$0.11	 SMA 37 mm ²
Df	Diodes Inc.	B260A-13-F	VF@Io= 700.0 mV VRRM= 60.0 V	1	\$0.10	 SMA 37 mm ²
L1	Coilcraft	XAL7070-222MEB	L= 2.2 uH 5.7 mOhm	1	\$1.19	 XAL7070 87 mm ²
M1	Texas Instruments	CSD19534Q5A	VdsMax= 100.0 V IdsMax= 50.0 Amps	1	\$0.28	 TRANS_NexFET_Q5A 55 mm ²
M2	Texas Instruments	CSD19534Q5A	VdsMax= 100.0 V IdsMax= 50.0 Amps	1	\$0.28	 TRANS_NexFET_Q5A 55 mm ²
M3	Texas Instruments	CSD17578Q5A	VdsMax= 30.0 V IdsMax= 59.0 Amps	1	\$0.15	 TRANS_NexFET_Q5A 55 mm ²
M4	Texas Instruments	CSD17578Q5A	VdsMax= 30.0 V IdsMax= 59.0 Amps	1	\$0.15	 TRANS_NexFET_Q5A 55 mm ²
Rcomp	Vishay-Dale	CRCW04025K90FKED Series= CRCW..e3	Res= 5.9 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²

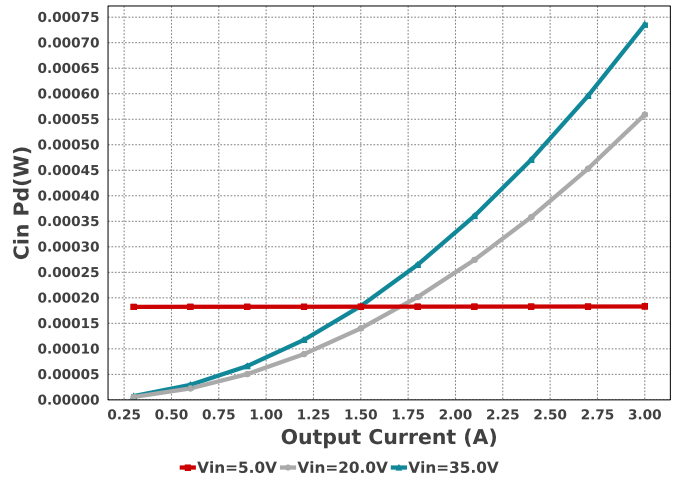
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Rcsg	Vishay-Dale	CRCW0402100RFKED Series= CRCW..e3	Res= 100.0 Ohm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rcsp	Vishay-Dale	CRCW0402100RFKED Series= CRCW..e3	Res= 100.0 Ohm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rf	Vishay-Dale	CRCW040210R0FKED Series= CRCW..e3	Res= 10.0 Ohm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rfbb	Yageo	RC0603FR-0720KL Series= ?	Res= 20.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Rfbt	Yageo	RC0603FR-07357KL Series= ?	Res= 357.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Rmode	Yageo	AC0402FR-07205KL Series= ?	Res= 205.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rpg	Vishay-Dale	CRCW040210K0FKED Series= CRCW..e3	Res= 10.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rsense	Rohm	PMR100HZPFU10L0 Series= PMR100	Res= 10.0 mOhm Power= 2.0 W Tolerance= 1.0%	1	\$0.19	2512 43 mm ²
Rt	Vishay-Dale	CRCW040213K0FKED Series= CRCW..e3	Res= 13.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Ruvb	Vishay-Dale	CRCW040293K1FKED Series= CRCW..e3	Res= 93.1 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rvvt	Vishay-Dale	CRCW0402249KFED Series= CRCW..e3	Res= 249.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rvisns	Vishay-Dale	CRCW06032K00FKEA Series= CRCW..e3	Res= 2.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
U1	Texas Instruments	LM5176PWPR	Switcher	1	\$3.33	PWP0028C_N 98 mm ²



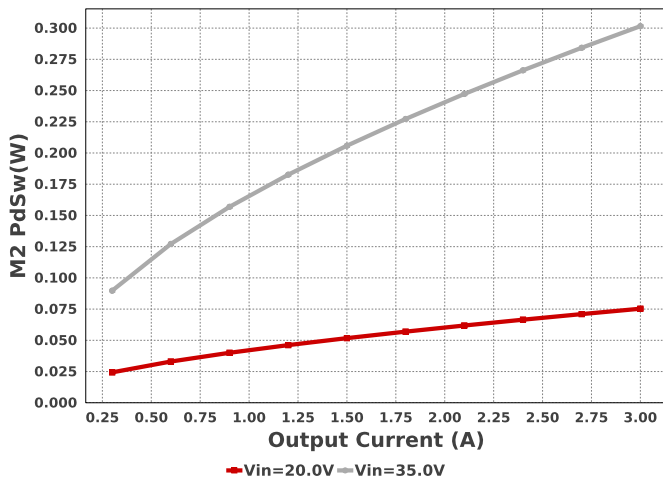
Cout IRMS



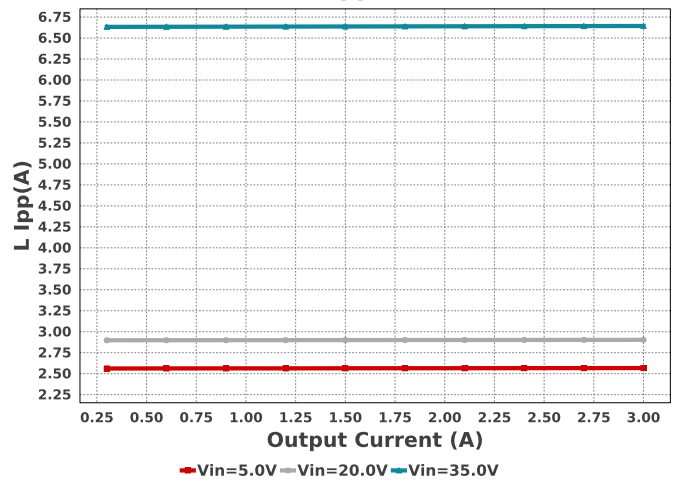
Cin Pd



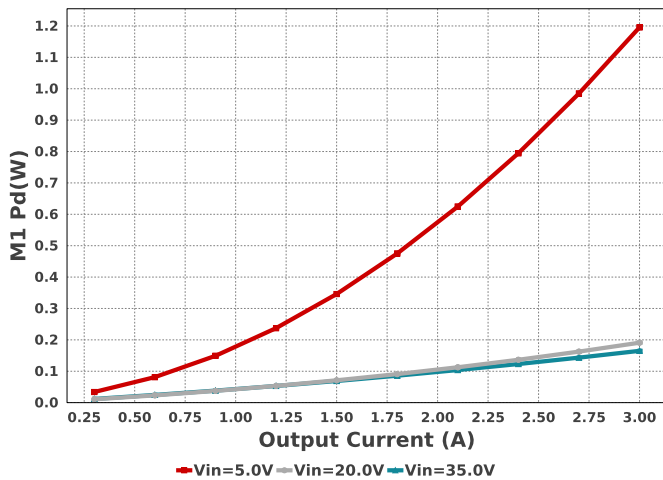
M2 PdSw



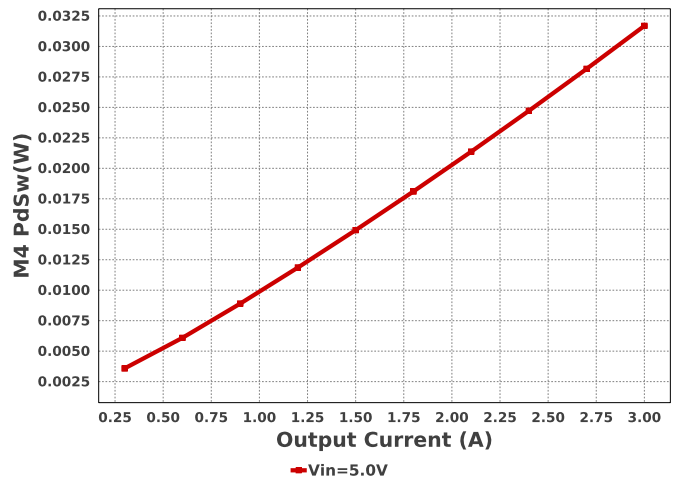
L Ipp

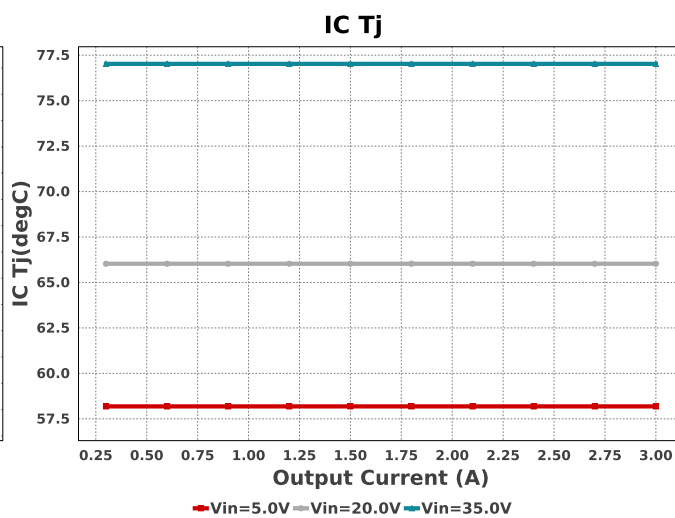
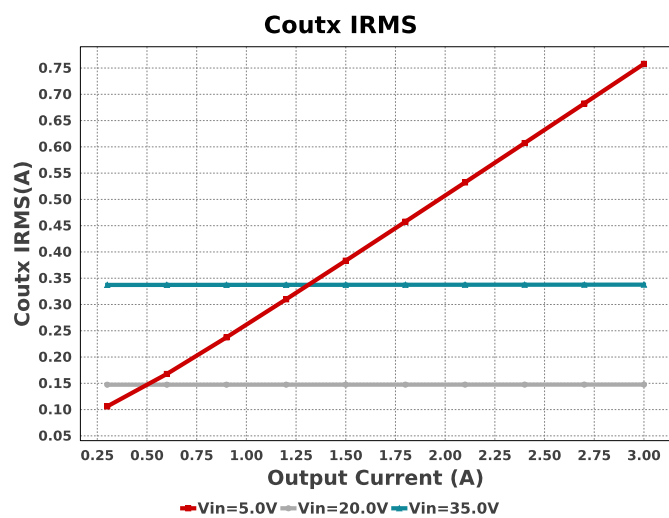
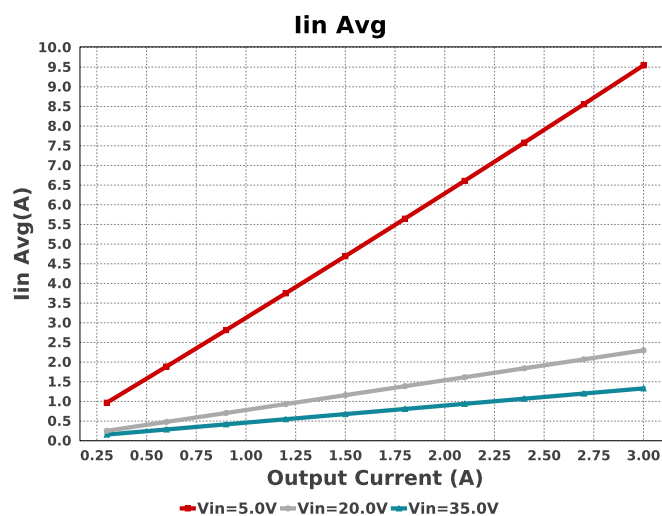
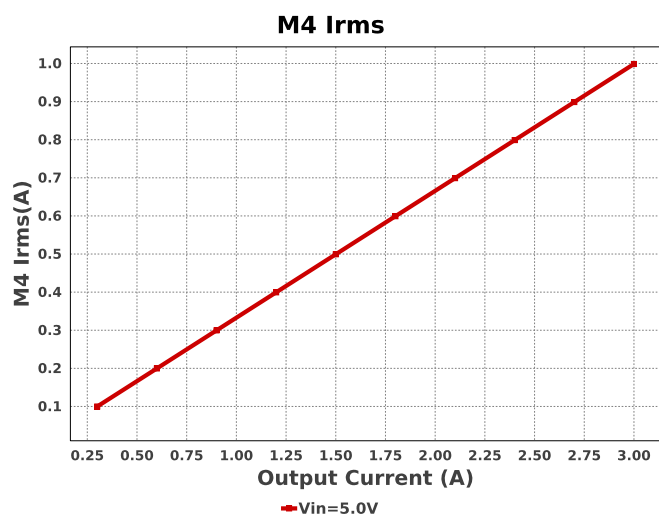
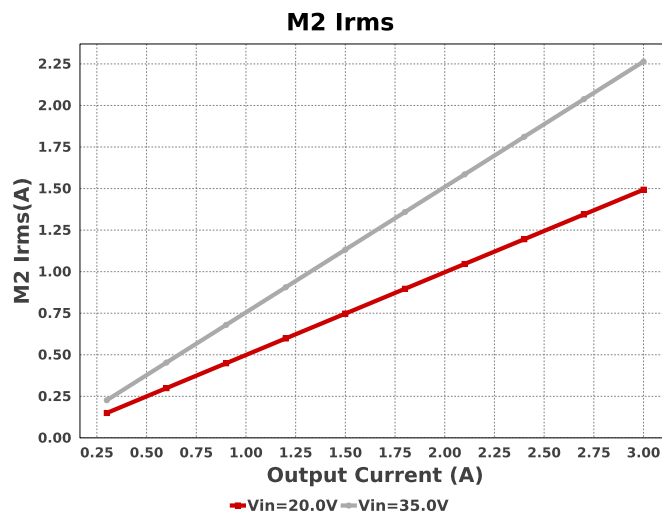
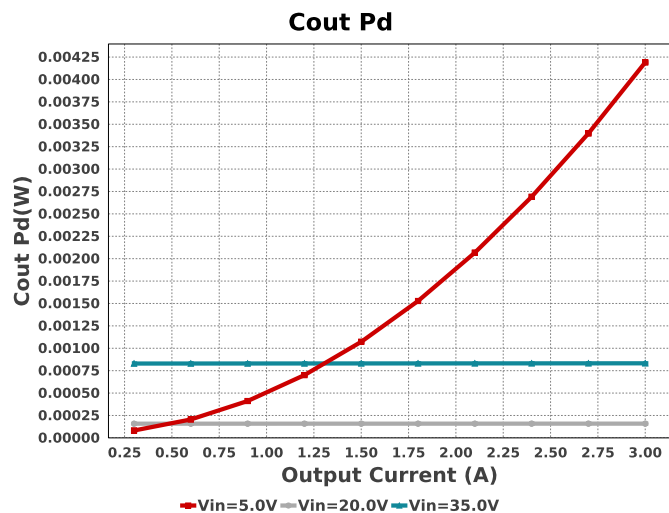


M1 Pd

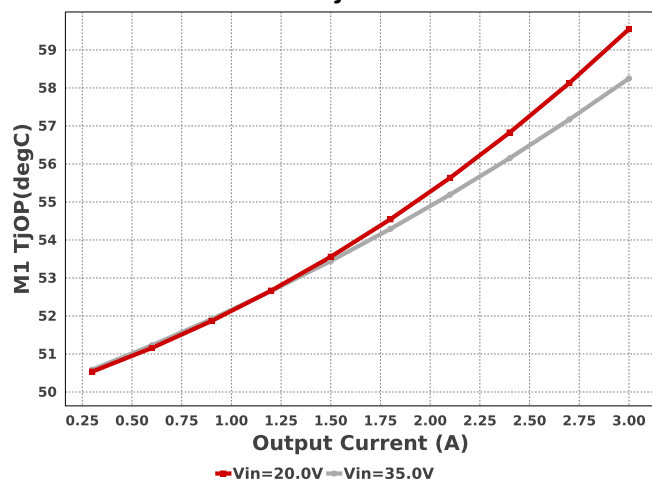


M4 PdSw

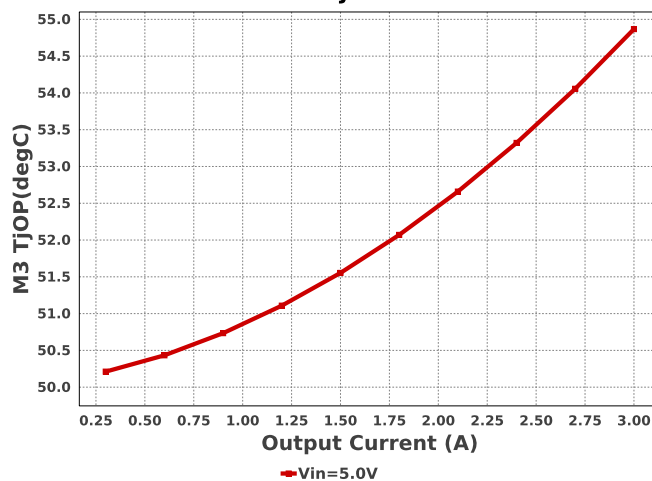




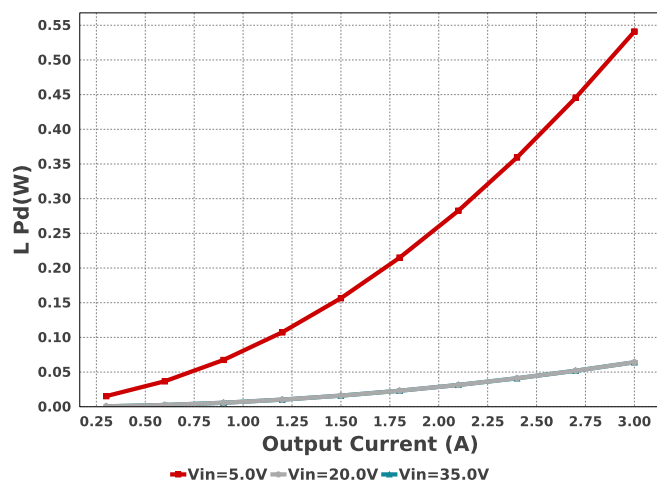
M1 TjOP



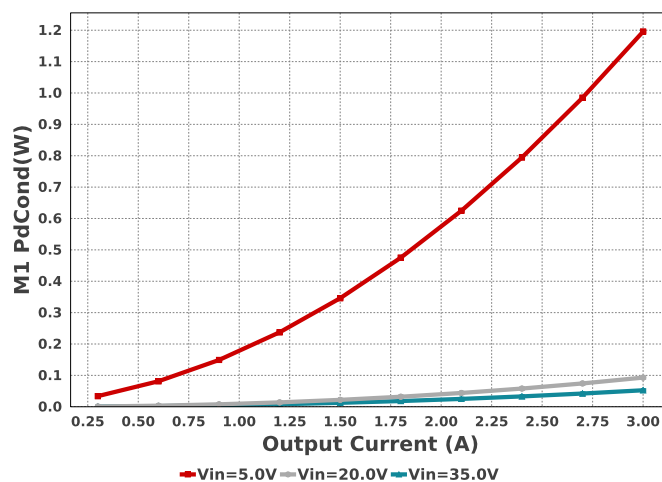
M3 TjOP



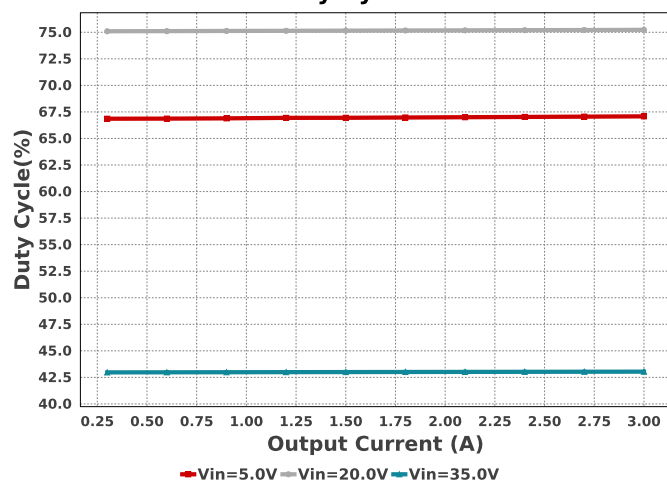
L Pd



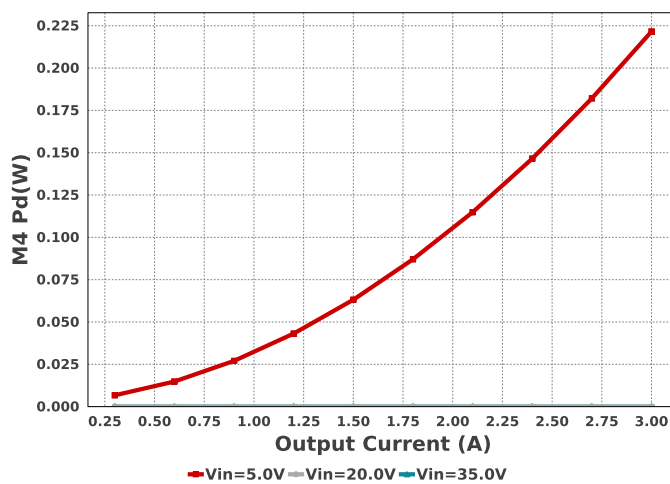
M1 PdCond



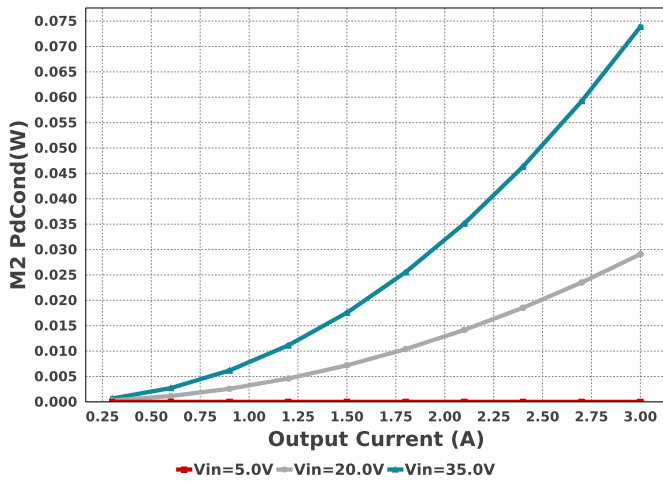
Duty Cycle



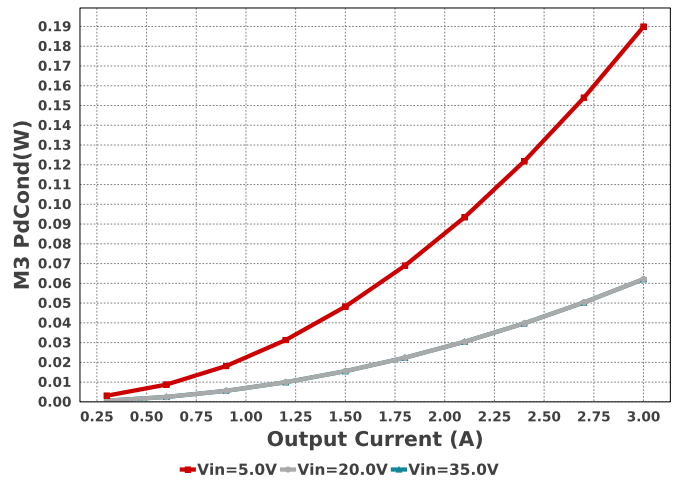
M4 Pd



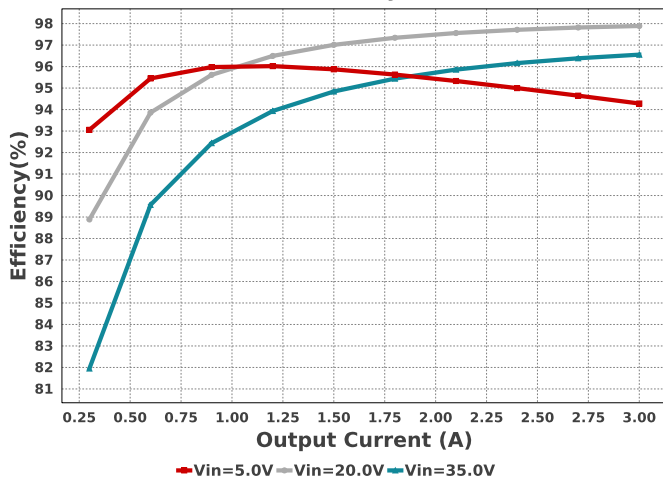
M2 PdCond



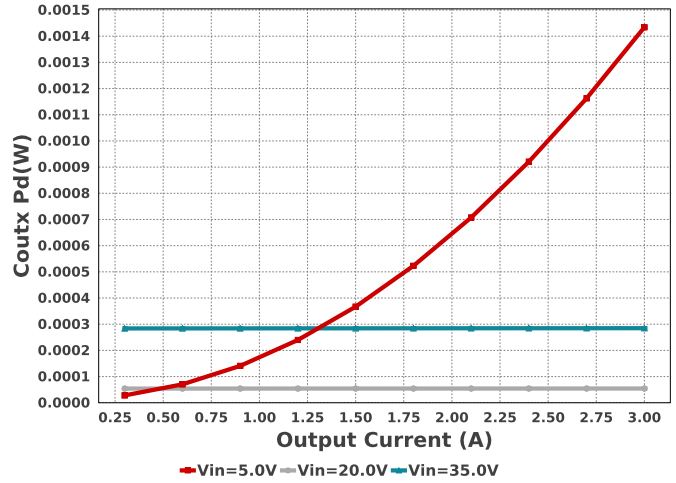
M3 PdCond



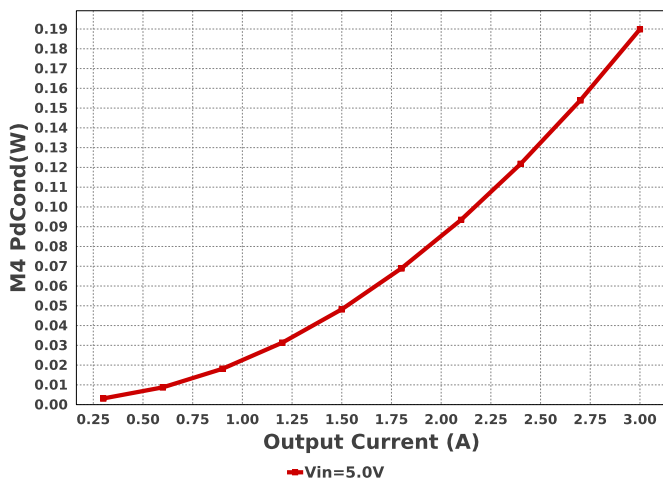
Efficiency



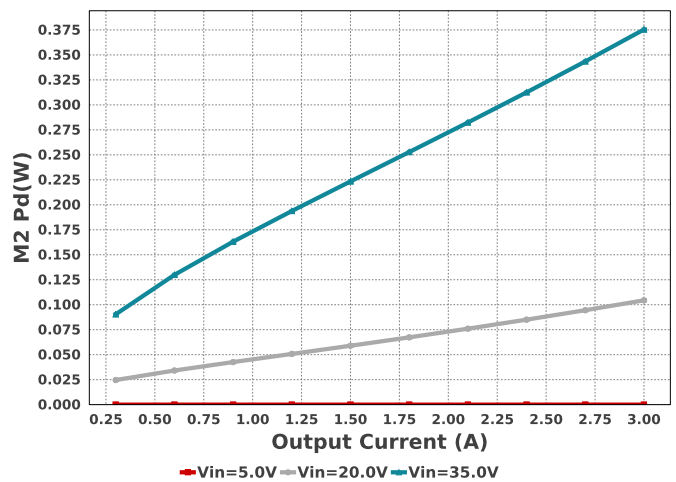
Coutx Pd

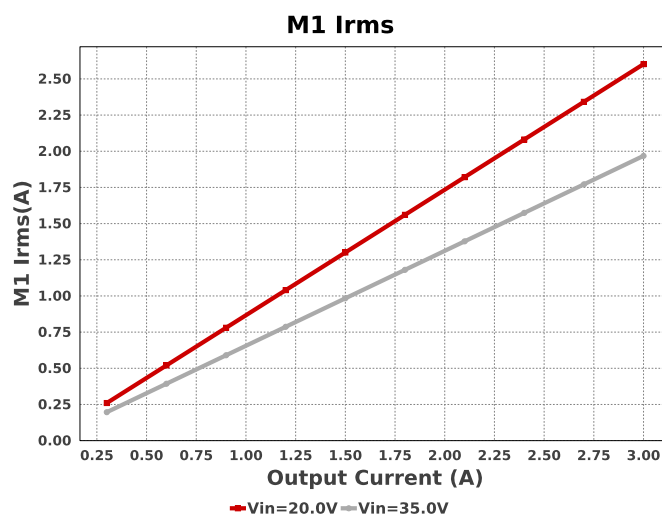
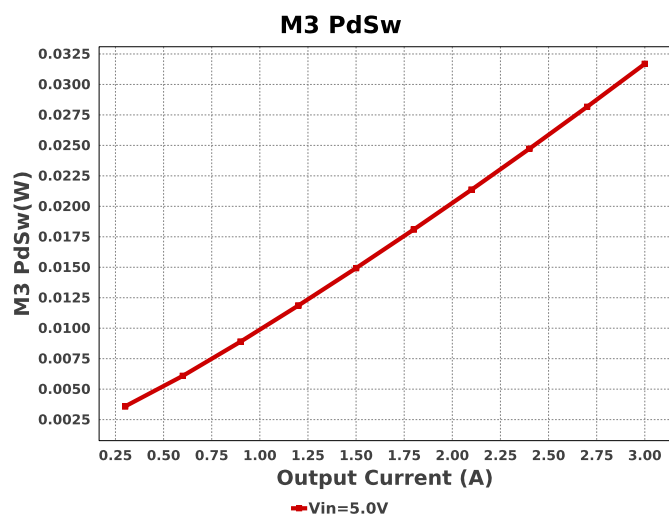
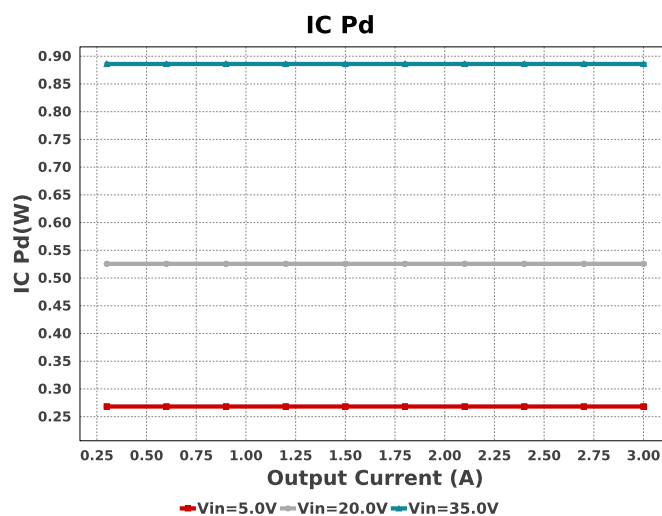
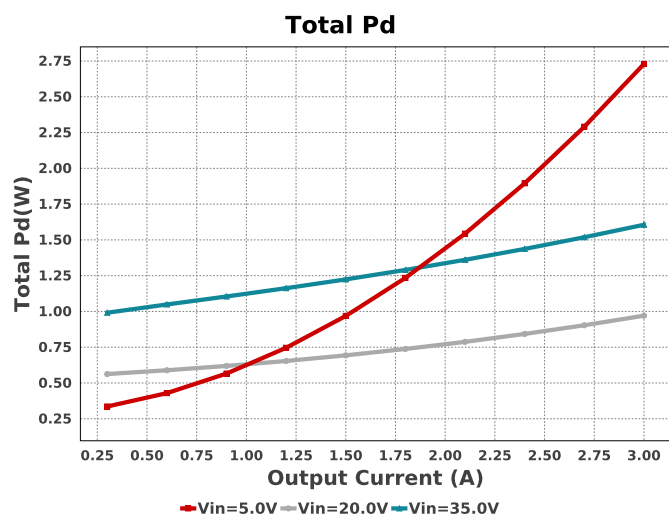
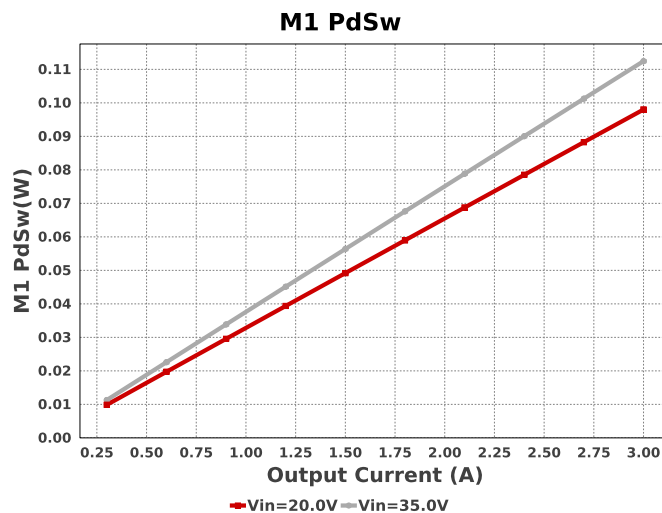
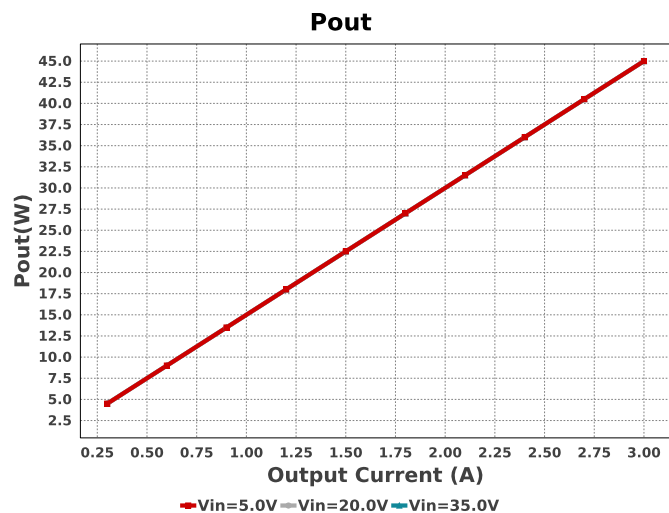


M4 PdCond

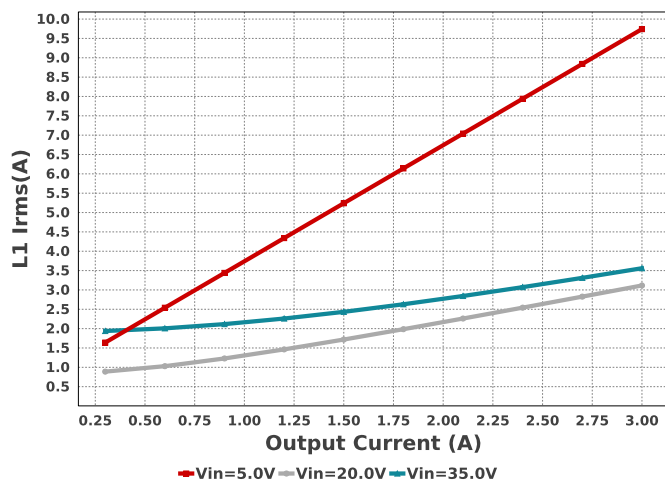


M2 Pd

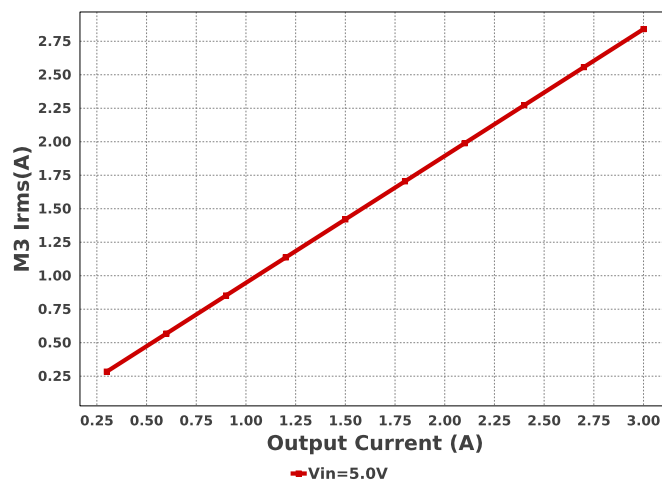




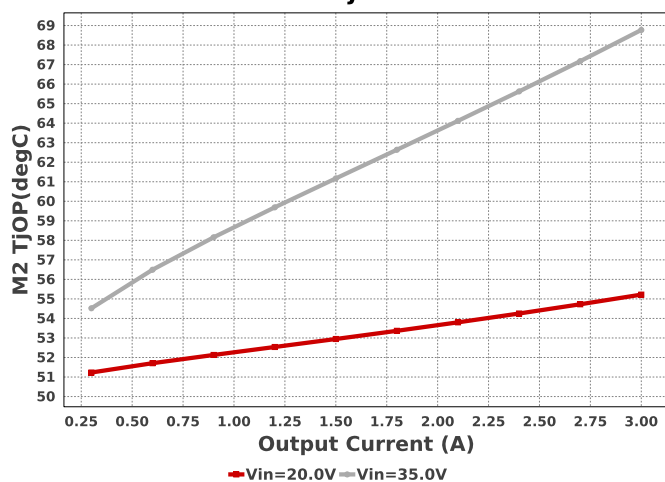
L1 Irms



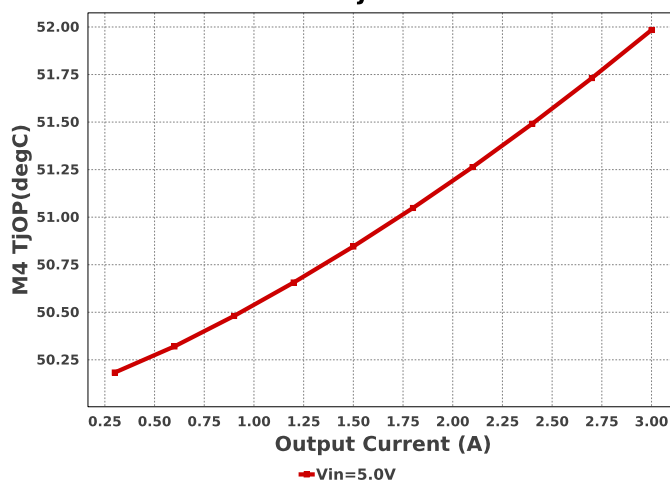
M3 Irms



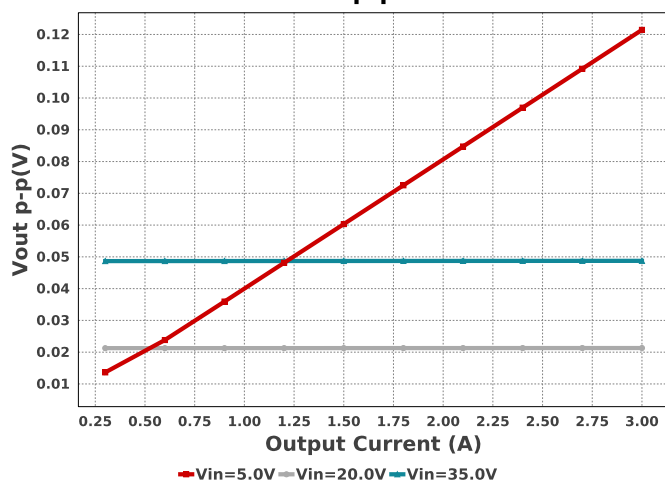
M2 TjOP



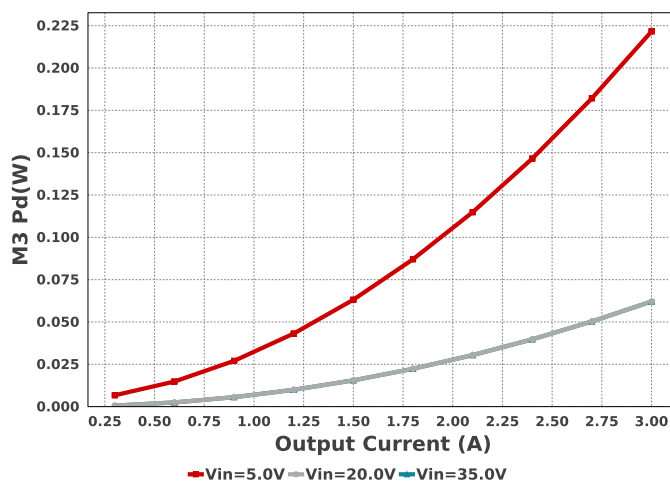
M4 TjOP

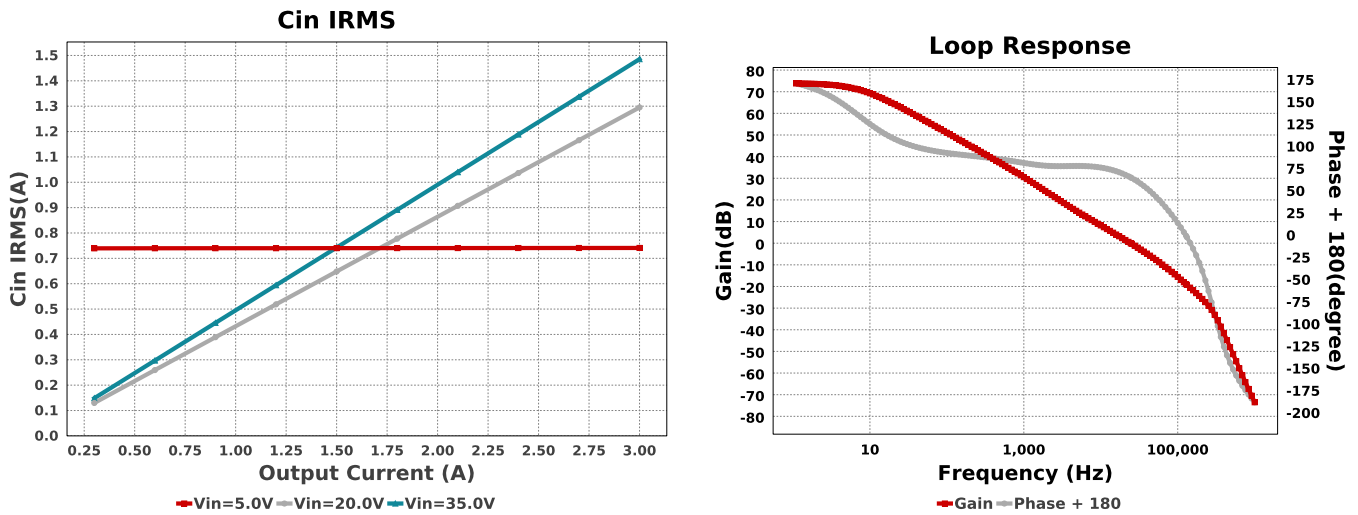


Vout p-p



M3 Pd





Operating Values

#	Name	Value	Category	Description
1.	BOM Count	55		Total Design BOM count
2.	Total BOM	\$10.402		Total BOM Cost
3.	Cin IRMS	1.485 A	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	735.47 μ W	Capacitor	Input capacitor power dissipation
5.	Cout IRMS	1.58 A	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	832.25 μ W	Capacitor	Output capacitor power dissipation
7.	Coutx IRMS	337.765 mA	Capacitor	Output capacitor_x RMS ripple current
8.	Coutx Pd	284.8 μ W	Capacitor	Output capacitor_x power loss
9.	IC Pd	885.99 mW	IC	IC power dissipation
10.	IC Tj	77.023 degC	IC	IC junction temperature
11.	IC Tolerance	0.0 V	IC	IC Feedback Tolerance
12.	ICThetaJA	30.5 degC/W	IC	IC junction-to-ambient thermal resistance
13.	Iin Avg	1.328 A	IC	Average input current
14.	L Ipp	6.644 A	Inductor	Peak-to-peak inductor ripple current
15.	L Pd	64.125 mW	Inductor	Inductor power dissipation
16.	L1 Irms	3.561 A	Inductor	Inductor ripple current
17.	M1 Irms	1.968 A	Mosfet	MOSFET RMS ripple current
18.	M1 Pd	202.85 mW	Mosfet	MOSFET power dissipation
19.	M1 PdCond	53.875 mW	Mosfet	M1 MOSFET conduction losses
20.	M1 PdSw	148.98 mW	Mosfet	M1 MOSFET switching losses
21.	M1 Rdson	12.6 mOhm	Mosfet	Drain-Source On-resistance
22.	M1 ThetaJA	50.0 degC/W	Mosfet	MOSFET junction-to-ambient thermal resistance
23.	M1 TjOP	60.143 degC	Mosfet	MOSFET junction temperature
24.	M2 Irms	2.264 A	Mosfet	MOSFET RMS ripple current
25.	M2 Pd	222.83 mW	Mosfet	MOSFET power dissipation
26.	M2 PdCond	69.286 mW	Mosfet	M2 MOSFET conduction losses
27.	M2 PdSw	153.55 mW	Mosfet	M2 MOSFET switching losses
28.	M2 Rdson	12.6 mOhm	Mosfet	Drain-Source On-resistance
29.	M2 ThetaJA	50.0 degC/W	Mosfet	MOSFET junction-to-ambient thermal resistance
30.	M2 TjOP	61.142 degC	Mosfet	MOSFET junction temperature
31.	M3 Pd	62.1 mW	Mosfet	M3 MOSFET total power dissipation
32.	M3 PdCond	62.1 mW	Mosfet	M3 MOSFET conduction losses
33.	M4 Pd	100.0 μ W	Mosfet	M4 MOSFET total power dissipation
34.	Cin Pd	735.47 μ W	Power	Input capacitor power dissipation
35.	Cout Pd	832.25 μ W	Power	Output capacitor power dissipation
36.	Coutx Pd	284.8 μ W	Power	Output capacitor_x power loss
37.	IC Pd	885.99 mW	Power	IC power dissipation
38.	L Pd	64.125 mW	Power	Inductor power dissipation
39.	M1 Pd	202.85 mW	Power	MOSFET power dissipation
40.	M1 PdCond	53.875 mW	Power	M1 MOSFET conduction losses
41.	M1 PdSw	148.98 mW	Power	M1 MOSFET switching losses
42.	M2 Pd	222.83 mW	Power	MOSFET power dissipation
43.	M2 PdCond	69.286 mW	Power	M2 MOSFET conduction losses
44.	M2 PdSw	153.55 mW	Power	M2 MOSFET switching losses
45.	M3 Pd	62.1 mW	Power	M3 MOSFET total power dissipation
46.	M3 PdCond	62.1 mW	Power	M3 MOSFET conduction losses
47.	M4 Pd	100.0 μ W	Power	M4 MOSFET total power dissipation
48.	Rsense Pd	51.264 mW	Power	LED Current Rsns Power Dissipation
49.	Total Pd	1.491 W	Power	Total Power Dissipation
50.	Rsense Pd	51.264 mW	Resistor	LED Current Rsns Power Dissipation
51.	Cross Freq	24.495 kHz	System Information	Bode plot crossover frequency

#	Name	Value	Category	Description
52.	Duty Cycle	43.039 %	System Information	Duty cycle
53.	Efficiency	96.793 %	System Information	Steady state efficiency
54.	FootPrint	986.0 mm ²	System Information	Total Foot Print Area of BOM components
55.	Frequency	588.928 kHz	System Information	Switching frequency
56.	Gain Marg	-18.879 dB	System Information	Bode Plot Gain Margin
57.	Iout	3.0 A	System Information	Iout operating point
58.	Low Freq Gain	73.823 dB	System Information	Gain at 1Hz
59.	Mode	CCM	System Information	Conduction Mode
60.	Operating Topology	Buck	System Information	The current operating topology of the device
61.	Phase Marg	65.492 deg	System Information	Bode Plot Phase Margin
62.	Pout	45.0 W	System Information	Total output power
63.	SW Ipk	0.0 A	System Information	Peak switch current
64.	Vin	35.0 V	System Information	Vin operating point
65.	Vout	15.0 V	System Information	Operational Output Voltage
66.	Vout Actual	15.08 V	System Information	Vout Actual calculated based on selected voltage divider resistors
67.	Vout Tolerance	1.913 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
68.	Vout p-p	48.745 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	3.0	Maximum Output Current
SoftStart	2.0 ms	Soft Start Time (ms)
VinMax	35.0	Maximum input voltage
VinMin	5.0	Minimum input voltage
Vout	15.0	Output Voltage
base_pn	LM5176	Base Product Number
source	DC	Input Source Type
Ta	50.0	Ambient temperature
UserFsw	588.928 k	Customer Selected Frequency

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

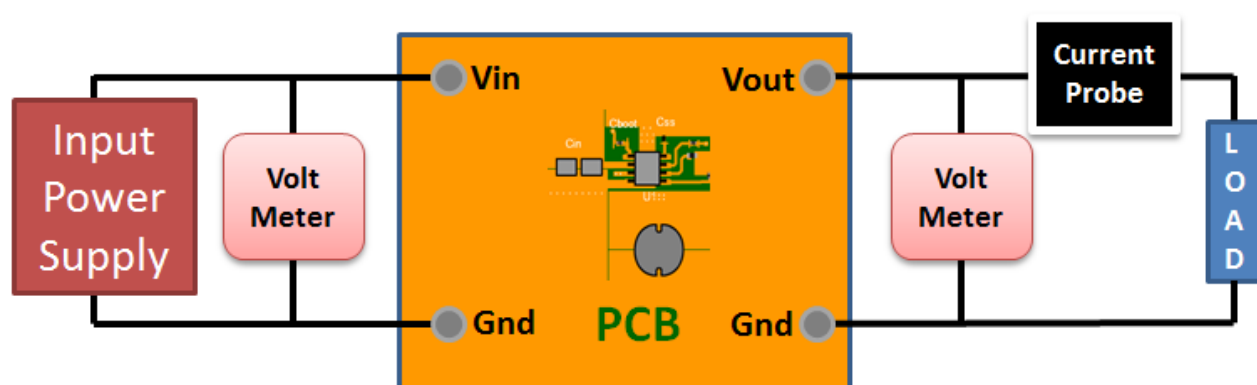
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 5.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Tip: Snubbers and/or gate resistors may be required to limit the SW1,2 node switching spikes below the IC and FET abs max ratings.
2. Tip: Slope Capacitor: smaller slope capacitors provide better transition region behavior.
3. Master key : F9884934EAC7C183[v1]
4. **LM5176** Product Folder : <http://www.ti.com/product/LM5176> : contains the data sheet and other resources.

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